NSCBC Subsonic Outlet Boundary Conditions

OpenFOAM v2412 (port) — User and Theory Manual

Contents

1 Overview

This library provides a partially reflecting (weakly non-reflecting) subsonic *outlet* for compressible reacting flow, implemented as three coupled boundary conditions — one per transported field that needs a characteristic treatment:

Type name	Applied to	Role
`pressureOutletNSCBC`	pressure p	acoustic relaxation to p_∞
`velocityOutletNSCBC`	velocity $\mathbf{U}$	acoustic + outgoing convection
`temperatureOutletNSCBC`	temperature T	entropy + acoustic transport

Species mass fractions Y_k at the outlet use OpenFOAM's built-in `advective` condition (Section 4.2). The three conditions share the same parameters (`pInf`, `etaAc`, `lInf`, `fieldInf`) and must be applied together on the same patch.

Scope. The implementation is the subsonic-outlet case of the NSCBC framework of Poinsot & Lele [1] with the Rudy–Strikwerda [3] pressure relaxation, evaluated with the variable-property thermodynamics of Baum, Poinsot & Thévenin [2] under a **frozen-composition** assumption ($\partial Y_k/\partial n \approx 0$). Composition derivatives in the wave amplitudes (the full Baum et al. multi-component extension) are *not* included. Within that scope the formulation has been verified term-by-term against both papers; this manual documents that mapping.

2 Theory

2.1 Characteristic waves (Poinsot & Lele)

On a boundary with outward normal n and normal velocity $u \equiv u_n$, the LODI (Local One-Dimensional Inviscid) analysis gives five characteristic waves with speeds

$$\lambda_1 = u - c, \qquad \lambda_{2,3,4} = u, \qquad \lambda_5 = u + c, \qquad c^2 = \gamma p/\rho \quad \text{(P\&L Eq. 18)}.$$

Their amplitude variations are (P&L Eqs. 19–23):

$$\mathcal{L}_1 = (u-c)\left(\frac{\partial p}{\partial n} - \rho c\frac{\partial u}{\partial n}\right), \qquad \text{incoming acoustic} \tag{1}$$

$$\mathcal{L}_2 = u\left(c^2\frac{\partial \rho}{\partial n} - \frac{\partial p}{\partial n}\right), \qquad \text{entropy} \tag{2}$$

$$\mathcal{L}_5 = (u+c)\left(\frac{\partial p}{\partial n} + \rho c\frac{\partial u}{\partial n}\right), \qquad \text{outgoing acoustic} \tag{3}$$

with $\mathcal{L}_3, \mathcal{L}_4 = u\,\partial u_{t_{1,2}}/\partial n$ the tangential-velocity waves (P&L Eqs. 21–22). All have units Pa/s.

Wave-numbering convention. This library uses the *Poinsot & Lele* ordering, in which $\mathcal{L}_2$ is the *entropy* wave and $\mathcal{L}_3, \mathcal{L}_4$ are transverse. Baum et al. [2] number the equal-speed waves differently ($\mathcal{L}_2, \mathcal{L}_3$ transverse, $\mathcal{L}_4$ entropy, then N_s species waves). The code and this manual follow P&L throughout.

2.2 LODI relations and gradient relations

At a subsonic outlet $0 < u < c$, so $\lambda_1 < 0$ (incoming) and $\lambda_{2..5} > 0$ (outgoing). The outgoing amplitudes $\mathcal{L}_2, \mathcal{L}_5$ are computed from one-sided *interior* gradients; the incoming $\mathcal{L}_1$ must be supplied (Section 2.3). The boundary fields are then advanced with the primitive LODI relations (P&L Eqs. 25, 26, 29):

$$\frac{\partial p}{\partial t} + \tfrac{1}{2}(\mathcal{L}_1 + \mathcal{L}_5) = 0, \tag{4}$$

$$\frac{\partial u}{\partial t} + \tfrac{1}{2\rho c}(\mathcal{L}_5 - \mathcal{L}_1) = 0, \tag{5}$$

$$\frac{\partial T}{\partial t} + \frac{T}{\rho c^2}\Big[-\mathcal{L}_2 + \tfrac{\gamma-1}{2}(\mathcal{L}_5 + \mathcal{L}_1)\Big] = 0, \tag{6}$$

with transverse $\partial u_t/\partial t + \mathcal{L}_{3,4} = 0$ (Eqs. 27–28). The associated boundary-normal gradient relations (P&L Eqs. 34–36) are used to build the Neumann part of each condition:

$$\frac{\partial p}{\partial n} = \tfrac{1}{2}\Big(\tfrac{\mathcal{L}_5}{u+c} + \tfrac{\mathcal{L}_1}{u-c}\Big), \quad \frac{\partial u}{\partial n} = \tfrac{1}{2\rho c}\Big(\tfrac{\mathcal{L}_5}{u+c} - \tfrac{\mathcal{L}_1}{u-c}\Big), \quad \frac{\partial T}{\partial n} = \frac{T}{\rho c^2}\Big[-\tfrac{\mathcal{L}_2}{u} + \tfrac{\gamma-1}{2}\Big(\tfrac{\mathcal{L}_5}{u+c} + \tfrac{\mathcal{L}_1}{u-c}\Big)\Big]. \tag{7}$$

2.3 Pressure relaxation

$\mathcal{L}_1$ enters from outside and cannot come from interior data. Setting $\mathcal{L}_1 = 0$ (perfectly non-reflecting) leaves the mean pressure unconstrained and ill-posed; following Rudy & Strikwerda [3] it is relaxed weakly toward a far-field pressure p_∞ (P&L Eq. 40, identical to Baum et al. Eq. 26):

$$\mathcal{L}_1 = K\,(p - p_\infty), \qquad K = \sigma\,(1 - \mathrm{Ma}^2)\,\frac{c}{L}, \tag{8}$$

(σ = `etaAc`, L = `lInf`). Small K approaches the non-reflecting limit; large K approaches a fixed-pressure outlet. A residual reflection coefficient is therefore unavoidable — the deliberate trade-off between non-reflectivity and well-posedness.

Note (local vs. maximum Mach number). Poinsot & Lele write Eq. (8) with the *maximum* Mach number of the flow; this library uses the *local* patch Mach number $\mathrm{Ma} = u/c$ = `aP/cP`, the now-common form. The two coincide for low-Mach outlets.

2.4 Variable thermodynamic properties (Baum et al.)

The only thermodynamic coefficients appearing in Eqs. (4)–(7) are γ, ρ, c (and R). Rather than treating them as constant (P&L), this library realises the advance of Baum et al. by evaluating them *face-by-face* from the live `fluidThermo` registry:

$$c = \sqrt{\gamma/\psi}, \qquad \gamma = c^2\psi, \qquad R = \frac{1}{\psi T},$$

where ψ = `thermo:psi` $= 1/(RT)$ is the compressibility. Because ψ and γ track local temperature and composition, c, the factor $(1 - \mathrm{Ma}^2)$ in K, and the LODI scaling $T/(\rho c^2) = 1/(\gamma\rho R)$ are all spatially correct even across a flame front.

2.5 Reduction of Baum et al. to Poinsot & Lele

Baum et al. [2] write the system in (ρ, T, u, v, w, Y_k) primitive variables with $N_s + 5$ waves; the temperature LODI is their Eq. 14,

$$\frac{\partial T}{\partial t} + \mathcal{L}_1 + \mathcal{L}_{N_s+5} + \textstyle\sum_{i=4}^{N_s+3} \mathcal{L}_i = 0.$$

Under frozen composition $\partial Y_k/\partial n \to 0$: (i) the N_s species waves vanish; (ii) the species-gradient corrections to the acoustic and entropy waves vanish. Re-expressing the remaining (ρ, T) waves in the (p, u) variables of P&L through the fixed-composition ideal-gas relations $c^2 = \gamma RT$ and $\mathrm{d}p = RT\,\mathrm{d}\rho + \rho R\,\mathrm{d}T$, Baum's Eq. 14 collapses term-by-term onto P&L Eq. 29 (6); the pressure and velocity equations likewise recover P&L Eqs. 25–26. **The frozen-composition outlet of Baum et al. is thus algebraically identical to the outlet of Poinsot & Lele in P&L wave notation**; the physics of Baum et al. survives only through the variable properties of the previous subsection.

3 Implementation: LODI as an OpenFOAM mixed (Robin) condition

Each field is implemented as a `mixedFvPatchField`, whose face value is

$$\varphi_f = w\,\varphi_{\mathrm{ref}} + (1-w)\big[\varphi_C + \Delta\,(\partial\varphi/\partial n)_{\mathrm{ref}}\big], \qquad \Delta = \tfrac{1}{\mathrm{deltaCoeffs}}, \quad \varphi_f - \varphi_C = \Delta\frac{\partial \varphi}{\partial n}, \tag{9}$$

with w = `valueFraction`, φ_{ref} = `refValue`, $(\partial\varphi/\partial n)_{\mathrm{ref}}$ = `refGrad`. The three coefficients are chosen so that (9) reproduces the discretised LODI relations. With $A \equiv K\Delta t/2$ and $B \equiv \frac{u+c}{2}\Delta t/\Delta$ (Euler / Crank–Nicolson branch):

Pressure. $w = \dfrac{1+A}{1+A+B}$, $\varphi_{\mathrm{ref}} = \dfrac{p^n + A\,p_\infty}{1+A}$, $(\partial p/\partial n)_{\mathrm{ref}} = -\rho c\,\frac{\partial u}{\partial n}$. Substituting into (9) and simplifying:

$$\frac{p_f - p^n}{\Delta t} + \underbrace{\tfrac{K}{2}(p-p_\infty)}_{\mathcal{L}_1/2} + \underbrace{\tfrac{u+c}{2}\Big(\frac{\partial p}{\partial n} + \rho c\,\frac{\partial u}{\partial n}\Big)}_{\mathcal{L}_5/2} = 0 \;\Rightarrow\; \frac{\partial p}{\partial t} + \tfrac{1}{2}(\mathcal{L}_1 + \mathcal{L}_5) = 0,$$

i.e. exactly P&L Eq. 25.

Normal velocity. $w = \dfrac{1}{1+B}$, $\varphi_{\mathrm{ref}} = u^n + \dfrac{K\Delta t}{2\rho c}(p-p_\infty)$, $(\partial u/\partial n)_{\mathrm{ref}} = -\dfrac{1}{\rho c}\frac{\partial p}{\partial n}$, which gives

$$\frac{\partial u}{\partial t} + \tfrac{1}{2\rho c}(\mathcal{L}_5 - \mathcal{L}_1) = 0 \qquad \text{(P\&L Eq. 26)}.$$

The minus sign in each `refGrad` is *required*: it supplies the $\rho c\,\partial u/\partial n$ part of $\mathcal{L}_5$ only after combining with the $(\varphi_f - \varphi_C)$ term of (9). Read in isolation a `refGrad` resembles the $u-c$ impedance, but the assembled result is the correct outgoing-wave LODI.

Temperature. The patch advects at u ($w = 1/(1+u\Delta t/\Delta)$) and carries the acoustic and entropy sources through `refValue`/`refGrad`; using $T/(\rho c^2) = 1/(\gamma\rho R)$ the assembly returns P&L Eq. 29 (6), and `refGrad` is the isentropic ($\mathcal{L}_2 = 0$) reduction of Eq. 36, $\partial T/\partial n = \frac{\gamma-1}{\gamma}\,\partial p/\partial n/(\rho R)$. In this v2412 port the entropy term $T/(\rho c^2)\mathcal{L}_2$ is carried by the convective operator (`valueFraction` at speed u) plus the pressure-gradient `refGrad`, rather than added explicitly to `refValue`; this avoids double-counting the entropy wave and is equivalent to Eq. 29.

Transverse velocity. The NSCBC source terms are projected onto the patch-normal direction via `fieldInf`; tangential components reduce to zero-gradient extrapolation, consistent with $\partial u_t/\partial t + \mathcal{L}_{3,4} = 0$ in the limit of vanishing normal-transverse gradients.

Time schemes. `Euler` and `CrankNicolson` use first-order LODI time integration (weights above). `backward` uses BDF2 time levels ($1 \to \frac{3}{2}$ in w and the two-level φ^n, φ^{n-1} in `refValue`) with the same spatial operators. Other `ddt` schemes are rejected at run time.

4 Usage

Register the libraries in `system/controlDict`:

```
libs
(
    "libpressureOutletNSCBCNew.so"
    "libvelocityOutletNSCBCNew.so"
    "libtemperatureOutletNSCBCNew.so"
);
```

Apply all three on the outlet patch (plus `advective` for species):

```
// 0/p
outlet { type pressureOutletNSCBC; pInf 101325; etaAc 0.25; lInf 0.05; fieldInf 1;
    value uniform 101325; }
// 0/U
outlet { type velocityOutletNSCBC; pInf 101325; etaAc 0.25; lInf 0.05; fieldInf (1 0 0)
    ; value uniform (0 0 0); }
// 0/T
outlet { type temperatureOutletNSCBC; pInf 101325; etaAc 0.25; lInf 0.05; fieldInf 1;
    value uniform 300; }
// 0/Yi (each species)
outlet { type advective; phi phi; value uniform 0; }
```

4.1 Parameters

Parameter	Description	Required	Default
`pInf`	Target far-field pressure p_∞ [Pa]	yes	—
`etaAc`	Relaxation coefficient σ in K	yes	0.25
`lInf`	Relaxation length scale L [m] (e.g. flame-to-outlet distance)	yes	—
`fieldInf`	Direction multiplier (scalar `1`; for $\mathbf{U}$, outward unit normal)	yes	—
`gamma`	Override γ; if omitted, taken from `fluidThermo`	no	from thermo
`phi`, `rho`, `p`, `U`	Field-name overrides	no	`phi`/`rho`/`p`/`U`

4.2 Species boundary condition

The species LODI at the outlet is pure advection, $\partial Y_k/\partial t + u\,\partial Y_k/\partial n = 0$, which is exactly what OpenFOAM's `advective` condition solves — use it for every Y_k. A `zeroGradient` condition is the $\partial Y_k/\partial n \to 0$ limiting case and is acceptable when composition gradients at the outlet are truly negligible.

4.3 Build

```
./Allwmake # builds the three libpressureOutletNSCBCNew.so etc.
```

Libraries are written to `$FOAM_USER_LIBBIN`.

4.4 Guidance

- Place the outlet where the flow is subsonic and composition gradients are small; the LODI approximation is exact only for boundary-normal waves, so oblique waves leave a small residual reflection.
- Increasing `etaAc` or decreasing `lInf` pulls the mean pressure to p_∞ more strongly but reflects more acoustic energy; $\sigma \approx 0.25$ (P&L §3.2) is a sound starting point.
- Keep `pInf` consistent with the value/internal field used to initialise p.

5 Validation (summary)

The formulation was assessed in one dimension: (i) a right-running Gaussian *acoustic* pulse and (ii) a convected Gaussian *entropy* pulse, both leaving the outlet with strongly reduced reflection relative to a baseline `fixedValue/zeroGradient` outlet (a weak residual consistent with the relaxation term); and (iii) a laminar premixed flame transported through the outlet, which crossed the boundary with its structure preserved and the pressure held near p_∞. The condition therefore performs within the limits of the LODI model on which it is based.

References

[1] T. J. Poinsot and S. K. Lele, *Boundary conditions for direct simulations of compressible viscous flows*, J. Comput. Phys. **101**:104–129, 1992.

[2] M. Baum, T. Poinsot, and D. Thévenin, *Accurate boundary conditions for multicomponent reactive flows*, J. Comput. Phys. **116**:247–261, 1994.

[3] D. H. Rudy and J. C. Strikwerda, *A nonreflecting outflow boundary condition for subsonic Navier–Stokes calculations*, J. Comput. Phys. **36**:55–70, 1980.